

Marco Maida

Senior Machine Learning Engineer

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I'm a software and machine learning engineer with over **ten years of experience** delivering systems across **industrial software**, **videogames**, **real-time systems**, **formal verification**, and **autonomous vehicles**.

At **Wayve**, I develop real-time inference systems for autonomous vehicles. My work covers model and runtime co-design, heterogeneous accelerators, numerical validation, and latency optimisation. I work primarily in **C++**, **Rust**, and **Python**, with an emphasis on measurable performance, correctness, and production constraints.

Professional Experience

Wayve Ltd

London, United Kingdom

May 2026
→ Present

- **Senior Machine Learning Engineer**

Co-design models and runtime architectures for real-time execution on resource-constrained automotive accelerators. Partition workloads across CPUs, GPUs, and NPUs; start model components as their sensor inputs become available; and maintain validation suites for numerical precision. Optimise end-to-end latency, throughput, and resource utilisation. (*C++*, *Python*).

Dec 2024
→ May 2026

- **Tech Lead, Inference Performance**

Led a team of four responsible for Wayve's on-board inference stack. Extended the platform across multiple inference frameworks and accelerators while supporting the same execution path in every vehicle and simulation environment. Owned latency, reliability, and technical delivery for the core inference component. (*C++*, *Rust*, *Python*).

Dec 2022
→ Dec 2024

- **Senior Software Engineer**

Joined a seven-person team building Wayve's second-generation on-board platform and shipping it to its first two vehicles. Built the initial real-time inference pipeline; Lidar, GNSS, and microphone stacks; data collection and upload features; and the vehicles' internal software network. (*ROS2*, *Linux*, *C++*, *Rust*, *Python*).

34BigThings

Turin, Italy

Aug 2016
→ Sep 2019

- **Software Engineer – Videogames**

Shipped five single-player and online multiplayer titles across Steam, PS4, Xbox One, Nintendo Switch, and mobile. Developed game infrastructure, AI, gameplay systems, and production tools in Unity and Unreal Engine. (*Unity3D*, *C#*, *Unreal Engine*, *C++*).

Maserati / Choralia

Turin, Italy

Sep 2016
→ May 2017

- **Software Engineer (Contractor) – Games and Simulation**

Delivered two B2B projects: a browser and mobile educational game, and a 3D product-visualisation tool. Led implementation and supervised an external artist. (*Unity3D*, *C#*, *JavaScript*).

R.O. srl

Jun 2013
→ Aug 2016

- **Software Engineer – Industrial Software**

Developed software for glass-processing factories covering order tracking, machine scheduling, waste reduction, and logistics. Managed four junior engineers during the third year. (*C*, *C++*, *C#*, *SQL*).

Education and Research

Jun 2022 → Aug 2022	Research Internship I worked on accelerating SAT solving using GPUs. (C++, CUDA).	Bloomberg LP
Sep 2019 → Oct 2022	PhD Student My research focused on formal verification and real-time systems, specifically verifying the timeliness of software systems. I published three papers and mentored three interns. (COQ, C, Rust).	Max Planck Institute
Nov 2019 → Jul 2022	Master's in Computer Science I specialized in real-time systems as part of a joint master's and PhD program.	Technische Universität Kaiserslautern
Sep 2016 → Jun 2019	Bachelor's in Computer Science I specialized in computability and formal methods.	Università degli Studi di Torino

Selected Publications

2025	Claycode: Stylable and Deformable 2D Scannable Codes ★ <i>Journal publication in Transactions of Graphics. Selected for CAF trailer.</i> We introduced Claycodes, a new type of visual code that breaks free from traditional rigid grids like QR codes. Claycodes encode data as a tree structure and allow full customizability and deformation, with real-time scanning capabilities. (https://arxiv.org/abs/2505.08666).	SIGGRAPH 2025
2022	From Intuition to Coq: A Case Study in Verified Response-Time Analysis, FIFO Scheduling We developed a formally verified response-time analysis for FIFO schedulers, challenging traditional pen-and-paper methods. (https://people.mpi-sws.org/~kbedarka/rtss22.pdf).	RTSS 2022
2021	Foundational Response-Time Analysis as Explainable Evidence of Timeliness ★ <i>Outstanding Paper Award.</i> I developed POET, a tool for formally verified worst-case scenario timing analysis. (https://drops.dagstuhl.de/opus/volltexte/2022/16336/pdf/LIPIcs-ECRTS-2022-19.pdf).	ECRTS 2022

Selected Open-Source Projects

2025	Claycode Inventor and lead contributor. Claycode includes a generator website, an Android scanner app, and a large amount of research code.	https://claycode.io
2020 → 2022	PROSA Main contributor to PROSA, one of the leading formally-verified frameworks in the real-time systems community.	https://gitlab.mpi-sws.org/RT-PROOFS/rt-proofs
2022	Treecode The precursor to Claycode, a novel 2D scannable code that encodes messages as unique trees.	www.maida.me/treecode
2021	POET First-ever implementation of a foundational response-time analysis tool, created as part of my first academic publication.	https://gitlab.mpi-sws.org/RT-PROOFS/POET
2018	Fast Mobile Cycle (FMC) Framework and Toolkit Developed a Unity3D framework that accelerates production-ready casual game creation, paired with a Python toolkit for bulk operations.	www.github.com/340openThings